

```
import numpy as np
```

```
np.eye(3)
```

```
array([[1., 0., 0.],  
       [0., 1., 0.],  
       [0., 0., 1.]])
```

```
np.sum([1,2])
```

```
np.int64(3)
```

```
np.max([2,6,5])
```

```
np.int64(6)
```

```
np.min([2,4,1])
```

```
np.int64(1)
```

```
np.median([7,6,8])
```

```
np.float64(7.0)
```

```
np.sqrt([2,3])
```

```
array([1.41421356, 1.73205081])
```

```
np.mean([2,6,5])
```

```
np.float64(4.333333333333333)
```

```
np.append([2,6,7],8)
```

```
array([2, 6, 7, 8])
```

```
np.size([3,8,9,4])
```

```
4
```

```
np.unique_values([2,3,2,4,2])
```

```
array([2, 3, 4])
```

```
np.sort([6,3,4,1,5])
```

```
array([1, 3, 4, 5, 6])
```

```
np.array([1, 2, 3, 4])
```

```
array([1, 2, 3, 4])
```

```
np.zeros((3, 4))
```

```
array([[0., 0., 0., 0.],  
       [0., 0., 0., 0.],  
       [0., 0., 0., 0.]])
```

```
np.ones((2, 3))
```

```
array([[1., 1., 1.],  
       [1., 1., 1.]])
```

```
np.empty((2, 2))
```

```
array([[9.49410977e-316, 0.00000000e+000],
       [0.00000000e+000, 0.00000000e+000]])
```

```
np.arange(0, 10, 2)
```

```
array([0, 2, 4, 6, 8])
```

```
np.linspace(0, 1, 5)
```

```
array([0. , 0.25, 0.5 , 0.75, 1.  ])
```

```
np.identity(3)
```

```
array([[1., 0., 0.],
       [0., 1., 0.],
       [0., 0., 1.]])
```

```
np.full((2, 3), 7)
```

```
array([[7, 7, 7],
       [7, 7, 7]])
```

```
np.random.rand(2, 3)
```

```
array([[0.0700639 , 0.84080375, 0.88058573],
       [0.62302825, 0.43354316, 0.76626464]])
```

```
np.random.randint(low=0, high=10, size=(2, 3))
```

```
array([[0, 9, 8],
       [4, 4, 3]])
```

```
a = np.array([1, 2, 3])
print(a.dtype)
```

```
int64
```

```
a = np.array([[1, 2, 3],
              [4, 5, 6]])
print(a.shape)
```

```
(2, 3)
```

```
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
c = np.hstack((a, b))
print(c)
```

```
[1 2 3 4 5 6]
```

```
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
c = np.vstack((a, b))
print(c)
```

```
[[1 2 3]
 [4 5 6]]
```

```
a = np.array([1, 2, 3, 4, 5, 6])
b = np.split(a, 2)
print(b)
```

```
[array([1, 2, 3]), array([4, 5, 6])]
```

```
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
c = np.concatenate((a, b))
print(c)
```

```
[1 2 3 4 5 6]
```

```
a = np.array([1, 2, 3, 4])  
b = np.delete(a, 2)  
print(b)
```

```
[1 2 4]
```

```
a = np.array([1, 2, 3, 4])  
b = a.reshape(2, 2)  
print(b)
```

```
[[1 2]  
 [3 4]]
```

```
a = np.array([[1, 2, 3],  
              [4, 5, 6]])  
print(a.T)
```

```
[[1 4]  
 [2 5]  
 [3 6]]
```

```
a = np.array([1, 2, 3, 4, 5, 6])  
  
b = a.reshape(2, 3)  
  
print(b)
```

```
[[1 2 3]  
 [4 5 6]]
```

```
import numpy as np  
a = np.array([1, 2, 3, 4])  
b = np.array([5, 6, 7, 8])  
c = np.array([a, b])  
print(c)  
row_sum = np.sum(c, axis=1)  
print(row_sum)  
col_median = np.median(c, axis=0)  
print(col_median)
```

```
[[1 2 3 4]  
 [5 6 7 8]]  
[10 26]  
[3. 4. 5. 6.]
```

```
print(np.ndim(c))
```

```
2
```

```
import numpy as np  
a = np.random.randint(1, 100, 12)  
print(a)  
b = a.reshape(3, 4)  
print(b)
```

```
import numpy as np  
a = np.random.randint(1, 100, 12)  
print(a)  
b = a.reshape(3, 4)  
print(b)
```

```
[19 93 72 30 83 62 78 99 86 83 11 64]  
[[19 93 72 30]  
 [83 62 78 99]  
 [86 83 11 64]]
```

```
import numpy as np  
a = np.random.randint(1, 100, 6)  
b = np.random.randint(1, 100, 6)  
print(a)  
print(b)  
a2 = a.reshape(3, 2)  
b2 = b.reshape(3, 2)  
print(a2)
```

```
print(b2)
v = np.vstack((a2, b2))
print(v)
h = np.hstack((a2, b2))
print(h)
```

```
[81 49 73 12 38 84]
[73 80 85  1 33 40]
[[81 49]
 [73 12]
 [38 84]]
[[73 80]
 [85  1]
 [33 40]]
[[81 49]
 [73 12]
 [38 84]
 [73 80]
 [85  1]
 [33 40]]
[[81 49 73 80]
 [73 12 85  1]
 [38 84 33 40]]
```

```
import numpy as np
a = np.ones((3,3))
b = np.eye(3)
print(a)
print(b)
mul = a * b
print(mul)
print(np.sum(mul))
```

```
[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
3.0
```

```
import numpy as np
a = np.random.randint(1, 100, 6)
b = np.random.randint(1, 100, 6)
print("Array a:",a)
print("Array b:",b)
print("Merged and unique values:")
np.concatenate((a,b))
```

```
Array a: [ 7 87 85 97 92 41]
Array b: [10 27 54 47 11 27]
Merged and unique values:
array([ 7, 87, 85, 97, 92, 41, 10, 27, 54, 47, 11, 27])
```

```
import numpy as np
a=np.array([1,2,3,4])
b=a.copy()
print(b)
```

```
[1 2 3 4]
```

```
import numpy as np
a = np.array([1, 2, 3, 4, 5, 6])
b=a.view()
b[3]=20
print(a)
```

```
[ 1  2  3 20  5  6]
```

```
import numpy as np
```

```
a=np.array([1,2,3,4])  
print(a[2])
```

3

```
import numpy as np  
a=np.array([1,2,3,4])  
print(a[0:3])
```

[1 2 3]

```
import numpy as np  
a = np.array([10, 25, 5, 40, 30])  
print(a[a > 20])
```

[25 40 30]

```
import numpy as np  
a=np.array([1,2,3,4,5,6])  
print(a[[0,1,2]])
```

[1 2 3]

Start coding or [generate](#) with AI.